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String instrument

Abstract

The present invention relates to a string instrument having a new structure that can amplify resonance using an analog method. In the string instrument according to the present invention, since auxiliary strings are provided inside a main body, when a user strums strings to play the string instrument, sound generated by the strings is amplified as the auxiliary strings vibrate due to vibration of the strings. Therefore, there is an advantage in that the sound generated when playing the string instrument becomes more vibrant.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a 371 of international application of PCT application serial no. PCT/KR2023/003089, filed on Mar. 7, 2023 which claims the priority benefit of Korean application no. 10-2022-0032162, filed on Mar. 15, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

TECHNICAL FIELD

(2) The present invention relates to a string instrument having a new structure that can amplify resonance using an analog method.

BACKGROUND ART

(3) Generally, as illustrated in FIGS. **1** and **2**, an acoustic guitar used in accompaniment or the like includes a main body (**10**) having a space portion formed therein, a neck (**20**) provided to extend forward from one side of the main body (**10**) and have tuning pegs (**21**) provided at a front end portion, and a plurality of strings **30** having both ends fixed to the tuning pegs (**21**) and the main body (**10**).

(4) The main body (**10**) is formed in the shape of a case having an upper plate (**11**), a sound hole (**12**) is formed in the upper plate (**11**), and a bridge (**13**) is provided on the upper plate (**11**).

(5) The strings (**30**) have both ends fixed to the tuning pegs (**21**) and the bridge (**13**).

(6) Therefore, when a user strums the strings (**30**), the strings (**30**) vibrate and generate sound, and here, as the body vibrates and amplifies the sound, people around the acoustic guitar can listen to loud music.

(7) However, since the acoustic guitar generates sound simply by the resonance of the strings (**30**) provided outside the main body (**10**), there is a problem in that the sound feels weak.

(8) In order to address such a problem, an electric guitar that amplifies sound using a pickup coil has been developed, but since the electric guitar gives a different feeling of amplifying sound using an analog method, there is a need for a new method that can amplify resonance using an analog method and give a more vibrant feeling.

(9) The above problem identically occurs in all types of string instruments including strings (**30**) and a body having a space portion formed therein, such as a classic guitar, an ukulele, a violin, and a cello, in addition to the above-described acoustic guitar.

(10) Meanwhile, the strings (**30**) may be a type in which both ends are straight or a type in which a ball end (**41**) is provided at one end as illustrated in FIG. **3**.

DISCLOSURE

Technical Problem

(11) The present invention has been devised to address the above problem and is directed to providing a string instrument having a new structure that can amplify resonance using an analog method.

Technical Solution

(12) The present invention provides a string instrument including a main body (**10**) having a space portion formed therein and a sound hole (**12**) formed at one side thereof, a neck (**20**) provided to extend forward from the one side of the main body (**10**) and have a tuning peg (**21**) provided at a front end portion, and a plurality of strings (**30**) having both ends fixed to the tuning peg (**21**) and the main body (**10**), wherein a through-hole (**14**) is formed at the one side of the main body (**10**), and an auxiliary string (**40**) inserted into the main body (**10**) through the through-hole (**14**) is provided inside the main body (**10**).

(13) According to another feature of the present invention, the string instrument may further include a support member (**50**) coupled to the through-hole (**14**) and an auxiliary tuning peg (**61**)

provided at the other side of the main body (**10**), and the auxiliary string (**40**) may have both ends fixed to the support member (**50**) and the auxiliary tuning peg (**61**).
(14) According to yet another feature of the present invention, an auxiliary through-hole (**15**) may be formed at the other side of the main body (**10**), the string instrument may further include an auxiliary support member (**62**) coupled to the auxiliary through-hole (**15**), and the auxiliary tuning peg (**61**) may be provided at the auxiliary support member (**62**).
(15) According to still another feature of the present invention, the string instrument may further include a support frame (**90**) inserted into the main body (**10**) through the through-hole (**14**) and to which the auxiliary string (**40**) is fixed, and the support frame (**90**) may include a first support plate (**91**) detachably coupled to the through-hole (**14**), a second support plate (**92**) provided to be spaced apart from the first support plate (**91**), a support bar (**93**) configured to connect the first and second support plates (**91**, **92**), and an auxiliary tuning peg (**94**) provided at the first support plate (**91**) or the second support plate (**92**).
(16) According to still another feature of the present invention, the string instrument may further include a muting mechanism (**70**) provided at the main body (**10**) to selectively come in contact with the auxiliary string (**40**) and mute the auxiliary string (**40**).
(17) Still another feature of the present invention provides a string instrument including a main body (**10**) having a space portion formed therein, a neck (**20**) provided to extend forward from one side of the main body (**10**) and have a tuning peg (**21**) provided at a front end portion, and a plurality of strings (**30**) having both ends fixed to the tuning peg (**21**) and the main body (**10**), wherein rear end portions of the strings (**30**) pass through the main body (**10**) and extend to the inside of the main body (**10**), and the string instrument further includes a muting mechanism (**70**) provided at the main body (**10**) to selectively come in contact with the rear end portions of the strings (**30**) extending to the inside of the main body (**10**) and mute an auxiliary string (**40**).
(18) According to still another feature of the present invention, a bridge (**13**) may be provided on an upper plate (**11**) of the main body (**10**), communication holes (**11***a*, **13***a*) through which the rear end portions of the strings (**30**) pass may be formed in the upper plate (**11**) and the bridge (**13**), respectively, an auxiliary through-hole (**15**) may be formed in a peripheral surface of the main body (**10**), an auxiliary bridge (**16**) may be provided on the peripheral surface of the main body (**10**), and the rear end portions of the strings (**30**) may pass through the auxiliary through-hole (**15**) and be fixed to the auxiliary bridge (**16**).
Advantageous Effects
(19) Since a string instrument according to the present invention has auxiliary strings (**40**) provided inside a main body (**10**) thereof, when a user strums strings (**30**) to play the string instrument, sound generated by the strings (**30**) is amplified as the auxiliary strings (**40**) vibrate due to vibration of the strings (**30**).
(20) Therefore, there is an advantage in that the sound generated when playing the string instrument becomes more vibrant.

Description

DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a plan view illustrating one example of a conventional string instrument.
(2) FIG. **2** is a side cross-sectional view illustrating the one example of the conventional string instrument.
(3) FIG. **3** is a reference view illustrating a string used in the conventional string instrument.
(4) FIG. **4** is a plan cross-sectional view illustrating a string instrument according to the present invention.
(5) FIG. **5** is a plan cross-sectional view illustrating an enlarged view of a main part of the string

instrument according to the present invention.
(6) FIG. **6** is a side cross-sectional view illustrating the string instrument according to the present invention.
(7) FIGS. **7** to **9** are side cross-sectional views illustrating enlarged views of main parts of the string instrument according to the present invention.
(8) FIGS. **10** to **12** are side cross-sectional views illustrating modifications of the string instrument according to the present invention.
(9) FIG. **13** is a side cross-sectional view illustrating a string instrument according to a second embodiment of the present invention.
(10) FIG. **14** is a side cross-sectional view illustrating a support frame of the string instrument according to the second embodiment of the present invention.
(11) FIG. **15** is a side cross-sectional view illustrating an exploded state of the string instrument according to the second embodiment of the present invention.
(12) FIG. **16** is a side cross-sectional view illustrating a string instrument according to a third embodiment of the present invention.
(13) FIG. **17** is a side cross-sectional view illustrating an enlarged view of a main part of the string instrument according to the third embodiment of the present invention.
MODES OF THE INVENTION
(14) Hereinafter, the present invention will be described in detail with reference to the accompanying exemplary drawings.
(15) FIGS. **4** to **9** illustrate a string instrument according to the present invention and show a case in which the present invention is applied to a typical acoustic guitar having six strings **30**.
(16) According thereto, the string instrument is the same as the conventional string instrument in that the string instrument includes a main body **10** having a space portion formed therein, a neck **20** provided to extend forward from one side of the main body **10** and have tuning pegs **21** provided at a front end portion, and a plurality of strings **30** having both ends fixed to the tuning pegs **21** and the main body **10**.
(17) Here, the main body **10** is formed in the shape of a case having an upper plate **11**, a sound hole **12** is formed in the upper plate **11**, and a bridge **13** is provided on the upper plate **11**.
(18) The strings **30** have both ends fixed to the tuning pegs **21** and the bridge **13**.
(19) Also, according to the present invention, auxiliary strings **40** are provided inside the main body **10**.
(20) In more detail, a through-hole **14** is formed at one side of the main body **10**, and an auxiliary through-hole **15** is formed at the other side of the main body **10**.
(21) The through-hole **14** is formed in a front peripheral surface of the main body **10**, and the auxiliary through-hole **15** is formed in a rear peripheral surface of the main body **10**.
(22) Also, a support member **50** coupled to the through-hole **14** and an auxiliary tuning peg **61** provided at the other side of the main body **10** are further provided.
(23) As illustrated in FIGS. **5** to **7**, the support member **50** includes a support plate **51** having six fixing holes **51***a* formed to pass through front and rear surfaces thereof and a pair of extension plates **52** extending rearward from both sides of the rear surface of the support plate **51**.
(24) The auxiliary tuning peg **61** is provided at an auxiliary support member **62** coupled to the auxiliary through-hole **15**.
(25) The auxiliary support member **62** includes an auxiliary support plate **62***a* coupled to the auxiliary through-hole **15** and an auxiliary extension plate **62***b* extending forward from the auxiliary support plate **62***a*.
(26) As illustrated in FIGS. **4** and **8**, the auxiliary tuning peg **61** is provided as six auxiliary tuning pegs **61**, the auxiliary tuning pegs **61** are fixed to the auxiliary extension plate **62***b*, and operating levers **61***a* for operating the auxiliary tuning pegs **61** extend to the outside of the main body **10**.
(27) The type of string having a ball end **41** provided at a front end portion is used as the auxiliary

strings **40**, six auxiliary strings **40** are provided to correspond to the strings **30**, and each auxiliary string **40** has a front end portion coupled to the support member **50** so that the ball end **41** is caught in the fixing hole **51***a* of the support member **50** and a rear end portion coupled to the auxiliary tuning peg **61**.

(28) Therefore, when the rear end portions of the auxiliary strings **40** are allowed to pass through the fixing holes **51***a* and be coupled to the auxiliary tuning pegs **61** at the front of the support member **50**, and then the operating levers **61***a* of the auxiliary tuning pegs **61** are operated to wind the auxiliary strings **40** around the auxiliary tuning pegs **61**, the ball ends **41** of the auxiliary strings **40** are caught in the fixing holes **51***a* and fixed.

(29) Also, when the auxiliary tuning pegs **61** are further tightened, as the auxiliary strings **40** are pulled taut, sound of each auxiliary string **40** is adjusted.

(30) Here, preferably, each auxiliary string **40** may be tuned to C, E, G, B, D, or F #.

(31) Also, a muting mechanism **70** configured to selectively come in contact with the auxiliary strings **40** and mute the auxiliary strings **40** is provided at the main body **10**.

(32) As illustrated in FIGS. **8** and **9**, the muting mechanism **70** includes a muting member **71** configured to selectively come in contact with the auxiliary strings **40** and mute the auxiliary strings **40**, a rotating bar **72** extending rearward from the muting member **71** and vertically rotatably coupled to a bracket **73** provided at the auxiliary extension plate **62***b*, an elastic member **74** connected to the rotating bar **72** to apply pressure thereto so that the muting member **71** comes in close contact with the auxiliary strings **40**, an extension cable **75** connected to the rotating bar **72**, and a footrest **76** connected to an end portion of the extension cable **75** and vertically rotatably coupled to a support **77**.

(33) The extension cable **75** includes an inner cable **75***a* having both ends connected to the rotating bar **72** and the footrest **76** and a housing **75***b* configured to surround an outer side of the inner cable **75***a*, and the inner cable **75***a* is configured to slide inside the housing **75***b*.

(34) Since the extension cable **75** is commonly used in a bicycle brake or the like, further detailed description thereof will be omitted.

(35) Therefore, at ordinary times, the muting member **71** is in close contact with the auxiliary strings **40** due to the elastic member **74** and the auxiliary strings **40** do not vibrate, and when a user steps on the footrest **76** and the footrest **76** moves downward, as the inner cable **75***a* is pulled, the rotating bar **72** rotates downward and accordingly, the muting member **71** is spaced apart from the auxiliary strings **40** such that the auxiliary strings **40** can vibrate.

(36) Since the string instrument configured as described above has the auxiliary strings **40** provided inside the main body **10**, when the user strums the strings **30** to play the string instrument, sound generated by the strings **30** is amplified as the auxiliary strings **40** vibrate due to vibration of the strings **30**.

(37) Therefore, there is an advantage in that the sound generated when playing the string instrument becomes more vibrant.

(38) Also, since the through-hole **14** is formed at the one side of the main body **10**, the support member **50** coupled to the through-hole **14** and the auxiliary tuning pegs **61** provided at the other side of the main body **10** are further provided, and the auxiliary strings **40** have both ends fixed to the support member **50** and the auxiliary tuning pegs **61**, there is an advantage in that the sound of each auxiliary string **40** can be adjusted by operating the auxiliary tuning pegs **61**.

(39) In particular, since the auxiliary through-hole **15** is formed at the other side of the main body **10**, the auxiliary support member **62** is coupled to the auxiliary through-hole **15**, and the auxiliary tuning pegs **61** are provided at the auxiliary support member **62**, there is an advantage in that the auxiliary tuning pegs **61** are easy to install.

(40) Also, since the muting mechanism **70** configured to selectively come in contact with the auxiliary strings **40** and mute the auxiliary strings **40** is further provided at the main body **10**, the user may adjust sound to be generated or not generated by the auxiliary strings **40** by operating the

muting mechanism **70** and adjusting the muting member **71** to be in close contact with the auxiliary strings **40** or spaced apart from the auxiliary strings **40**.
(41) Therefore, there is an advantage in that the user can freely adjust the tone of the string instrument while playing the string instrument.
(42) Although the case in which the type of string having the ball end **41** provided at one end is used as the auxiliary strings **40** has been described above as the case of the present embodiment, when the type of string not having the ball end **41** is used as the auxiliary strings **40**, as illustrated in FIG. **10**, hooks **53** are provided at the support member **50** so that the front end portions of the auxiliary strings **40** can be hung on the hooks **53** and fixed.
(43) Also, although the auxiliary tuning pegs **61** have been described above as being provided at the auxiliary support member **62**, the auxiliary tuning pegs **61** may be directly provided at the main body **10** as illustrated in FIG. **11**
(44) Also, although the muting mechanism **70** has been described above as being configured to be operated by the user stepping thereon by foot, as illustrated in FIGS. **11** and **12**, the rotating bar **72** connected to the muting member **71** may extend to the outside of the main body **10**, and the user may rotate the rotating bar **72** by hand and adjust the muting member **71** to be in close contact with the auxiliary strings **40** or spaced apart from the auxiliary strings **40**.
(45) Alternatively, an electric operating device operated by electricity may be used as the muting mechanism **70**.
(46) Also, the muting mechanism **70** may be configured to be fixed in a state in which the muting member **71** is spaced apart from the auxiliary strings **40**.
(47) Also, the positions of the through-hole **14** and the auxiliary through-hole **15** may be freely changed.
(48) Also, the extension cable **75** may be detachably coupled to the rotating bar **72** or the footrest **76**, and a middle portion of the extension cable **75** may be configured to be separable as necessary.
(49) Also, at sites where there is a possibility of contact between the auxiliary strings **40** and each part of the string instrument, flats may be installed or separate support rods may be installed for the flats or the support rods to support each part of the auxiliary strings **40**. In this way, the auxiliary strings **40** can be prevented from coming in contact with each part of the string instrument, and simultaneously, the range at which the auxiliary strings **40** vibrate, that is, the length of the auxiliary strings **40**, may be shortened to enhance the tone of the string instrument.
(50) FIGS. **13** to **15** illustrate a string instrument according to another embodiment of the present invention in which a through-hole **14** is formed at one side of the main body **10**, and a support frame **90** to which the auxiliary strings **40** are fixed is provided at the through-hole **14**.
(51) The support frame **90** includes a first support plate **91** detachably coupled to the through-hole **14**, a second support plate **92** provided to be spaced apart from the first support plate **91**, a support bar **93** configured to connect the first and second support plates **91** and **92**, and auxiliary tuning pegs **94** provided at the second support plate **92**.
(52) The first support plate **91** is detachably coupled to the through-hole **14** using a fixing screw (not illustrated).
(53) Also, fixing holes **91***a* to which the ball ends **41** of the auxiliary strings **40** are fixed are formed in the first support plate **91**.
(54) Therefore, both ends of the auxiliary strings **40** may be coupled to the fixing holes **91***a* of the support frame **90** and the auxiliary tuning pegs **94** in a state in which the support frame **90** is separated, the auxiliary tuning pegs **94** may be adjusted to tune each auxiliary string **40**, and then the support frame **90** may be inserted into the main body **10** through the through-hole **14** and fixed to complete the installation of the auxiliary strings **40**.
(55) In the string instrument configured as described above, since the auxiliary strings **40** are coupled to the support frame **90** detachably coupled to the through-hole **14**, the support frame **90** may be separated to install or replace the auxiliary strings **40**.

(56) Therefore, there is an advantage in that the auxiliary strings **40** are easy to install or replace.
(57) Although the case in which the auxiliary tuning pegs **94** are provided at the second support plate **92** has been described above as the case of the present embodiment, the fixing holes **91***a* may be formed in the second support plate **92**, and the auxiliary tuning pegs **94** may be provided at the first support plate **91**.
(58) Also, although the structure of the muting mechanism **70** is omitted in the case of the present embodiment, the muting mechanism **70** may be provided at one side of the support frame **90**.
(59) Also, although it is described that a single support frame **90** is installed in the present embodiment, the present invention is not limited thereto. Within a range that a storage space inside the string instrument allows, a plurality of support frames may be installed. For example, two support frames may be disposed to intersect each other in an X-shape inside the string instrument.
(60) Also, although not illustrated, the support frame **90** may be fixed to one or more of an upper plate, side plates, and a lower plate of the string instrument inside the string instrument. For example, the support frame **90** may be directly fixed to the upper plate **11** inside the string instrument, or a fixing brace structure fixed to the upper plate **11** may be disposed inside the string instrument. Thus, the support plates **91** and **92** of the support frame **90** may be fixed to the main body **10**. However, the present invention is not limited thereto, and an additional fixing device for fixing the support frame **90** to the main body may be disposed between the support frame **90** and a rear plate (a plate opposing the upper plate **11**) of the main body **10**, and the support frame **90** may be firmly fixed to the main body **10** by the additional fixing device and the fixing brace structure.
(61) Also, although not illustrated, a support rail (not illustrated) for facilitating the attachment and detachment of the support frame **90** may be further provided inside the main body **10**. For example, the support rail (not illustrated) may be fixed to the upper plate **11** and the rear plate, and the support frame **90** may be detachably and firmly fixed to the upper plate **11** and the rear plate through the rail.
(62) FIGS. **16** and **17** illustrate a string instrument according to yet another embodiment of the present invention in which the rear end portions of the strings **30** pass through the main body **10** and extend to the inside of the main body **10**.
(63) To this end, communication holes **11***a* and **13***a* through which the rear end portions of the strings **30** pass are formed in the upper plate **11** and the bridge **13**, respectively, and an auxiliary bridge **16** is provided on a peripheral surface of the main body **10**.
(64) Here, the type of string having the ball end **41** provided at a rear end portion is used as the strings **30**.
(65) Also, an auxiliary through-hole **15** is formed in the peripheral surface of the main body **10**, that is, a rear side surface of the main body **10**.
(66) The auxiliary bridge **16** has a plurality of fixing holes **16***a* through which the strings **30** pass formed therein and is coupled to the auxiliary through-hole **15**.
(67) A method of installing the strings **30** in the string instrument configured as described above is as follows.
(68) First, the front end portions of the strings **30** are coupled to the fixing holes **16***a* at the rear of the auxiliary bridge **16**, and then the strings **30** are allowed to extend to the outside of the main body **10** through the communication holes **11***a* and **13***a*.
(69) Then, when distal end portions of the strings **30** are coupled to the tuning pegs **21** and the tuning pegs **21** are tightened, as the strings **30** are pulled, the ball ends **41** are caught in the fixing holes **16***a* of the auxiliary bridge **16**.
(70) Then, after further tightening the tuning pegs **21** and tuning each string **30**, the installation of the strings **30** may be completed.
(71) In the string instrument configured as described above, since the rear end portions of the strings **30** extend to the inside of the main body **10**, there is an advantage in that the structure is simplified compared to the above-described embodiments in which the auxiliary strings **40** should

be separately installed.
DESCRIPTION OF REFERENCE NUMERALS
(72) **10**: main body **20**: neck **30**: string **40**: auxiliary string **50**: support member **61**: auxiliary tuning peg **62**: auxiliary support member **70**: muting mechanism **90**: support frame

Claims

1. A string instrument, comprising: a main body having a space portion formed therein and a sound hole formed at one side thereof; a neck provided to extend forward from the one side of the main body and have a tuning peg provided at a front end portion; and a plurality of strings having both ends fixed to the tuning peg and the main body, wherein a through-hole is formed at the one side of the main body, and an auxiliary string inserted into the main body through the through-hole is provided inside the main body.
2. The string instrument of claim 1, further comprising: a support member coupled to the through-hole; and an auxiliary tuning peg provided at the other side of the main body, wherein the auxiliary string has both ends fixed to the support member and the auxiliary tuning peg.
3. The string instrument of claim 2, wherein: an auxiliary through-hole is formed at the other side of the main body; the string instrument further comprises an auxiliary support member coupled to the auxiliary through-hole; and the auxiliary tuning peg is provided at the auxiliary support member.
4. The string instrument of claim 1, further comprising a support frame inserted into the main body through the through-hole and to which the auxiliary string is fixed, wherein the support frame includes a first support plate detachably coupled to the through-hole, a second support plate provided to be spaced apart from the first support plate, a support bar configured to connect the first and second support plates, and an auxiliary tuning peg provided at the first support plate or the second support plate.
5. A string instrument, comprising: a main body having a space portion formed therein; a neck provided to extend forward from one side of the main body and have a tuning peg provided at a front end portion; and a plurality of strings having both ends fixed to the tuning peg and the main body, wherein rear end portions of the strings pass through the main body and extend to the inside of the main body, and the string instrument further comprises a muting mechanism provided at the main body to selectively come in contact with the rear end portions of the strings extending to the inside of the main body and mute an auxiliary string, wherein a bridge is provided on an upper plate of the main body; communication holes through which the rear end portions of the strings pass are formed in the upper plate and the bridge, respectively; an auxiliary through-hole is formed in a peripheral surface of the main body; an auxiliary bridge is provided on the peripheral surface of the main body; and the rear end portions of the strings pass through the auxiliary through-hole and are fixed to the auxiliary bridge.